MATH3030 Coursework

Multivariate Statistics — Spring 2026

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1 Question 1

1.1 Exploratory Data Analysis

The UN dataset contains GDP per capita, life expectancy, and population for 141 countries at 12 evenly spaced time points from 1952 to 2007. No values are missing. Sample sizes per continent are uneven (52 African, 32 Asian, 30 European, 25 American, and 2 Oceanian), so any continent-level summary for Oceania is effectively a two-country average of Australia and New Zealand. We summarise the 70-year dynamics with continent-mean trajectories for each variable and a table of 1952 and 2007 medians.

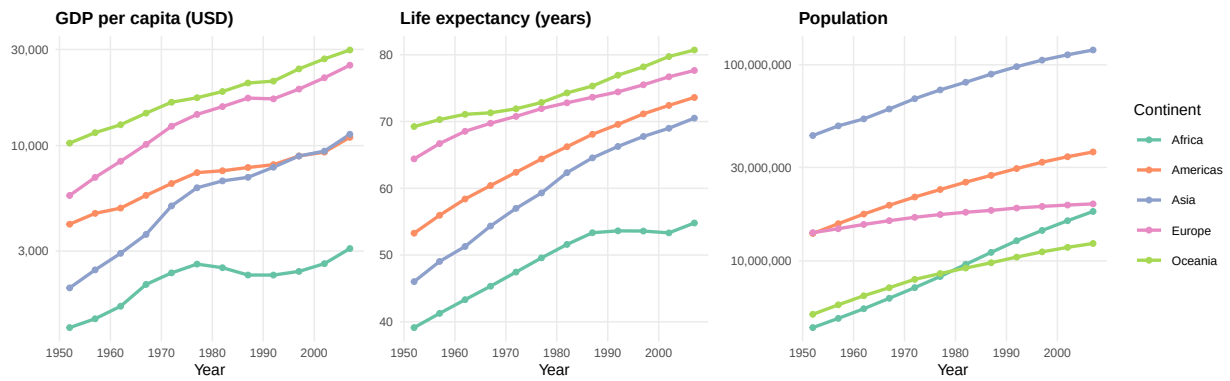

Figure 1: Continent-mean trajectories, 1952–2007. GDP and population are on log scales; life expectancy is linear.

Asia shows the steepest gains on both axes. Mean life expectancy rose from 44 to 72 years and median GDP per capita almost quadrupled, driven by a handful of export-led economies. Singapore's GDP per capita rose from \$2,315 to \$47,143 (20×), South Korea's by 23×, and Taiwan's by 24×. China dominates the population panel, rising from 556 million in 1952 to 1.32 billion in 2007. These multipliers are well above the Asian median GDP gain of 3.8×, so the continent mean hides a wide spread rather than uniform growth.

Africa's trajectory is much shallower. Median GDP rose only 1.5× over 55 years, and mean life expectancy, after rising from 39 to 53 years, plateaued from the 1990s onward. The plateau coincides with a broad health and conflict downturn in southern and central Africa. Zimbabwe's life expectancy fell from 48.5 to 43.5 years (the largest decline in the dataset), Swaziland, Zambia, and Lesotho show similar stagnation, and the Democratic Republic of the Congo is the only country whose GDP per capita ended below its 1952 level, at 36% of its original value. Africa's population grew faster than any other continent's, which compounds the GDP-growth shortfall on a per-capita basis.

Europe and Oceania remain firmly at the top throughout. Europe's GDP growth (\$5,143 to \$28,054, 5.5×) is comparable to Asia's in multiplier terms, but its life-expectancy gain of +13 years is about half of Asia's +28. This is structural rather than a lack of development. A population already close to its demographic ceiling has less room to gain years than one that is catching up. Both continents also evolve more smoothly than Asia or Africa, with no sharp deviations between adjacent five-year intervals.

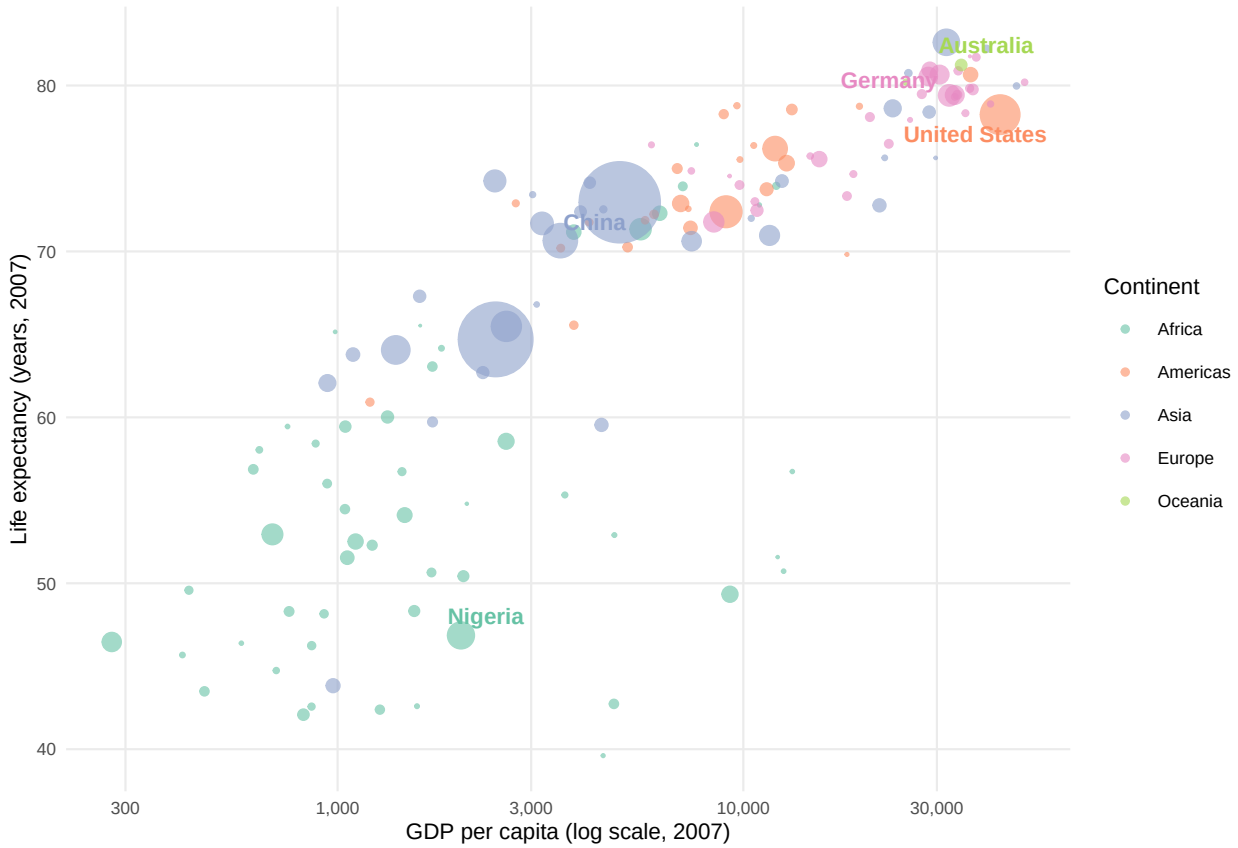

Figure 2: GDP per capita vs life expectancy in 2007. Bubble size is proportional to population.

The 2007 snapshot compresses the trajectories into a single cross-section and makes the joint structure visible. GDP and life expectancy are strongly but non-linearly related. The Pearson correlation on log(GDP) is 0.81, appreciably higher than on raw GDP (0.68), reflecting the diminishing marginal returns on life expectancy at high income. Continents separate cleanly, with Africa in the lower-left, Europe in the upper-right, and the Americas grouped in the middle. Asia is the outlier continent, spanning almost the full income range from India and Bangladesh at the low end to Japan and Singapore at the top, with China positioned between the two modes.

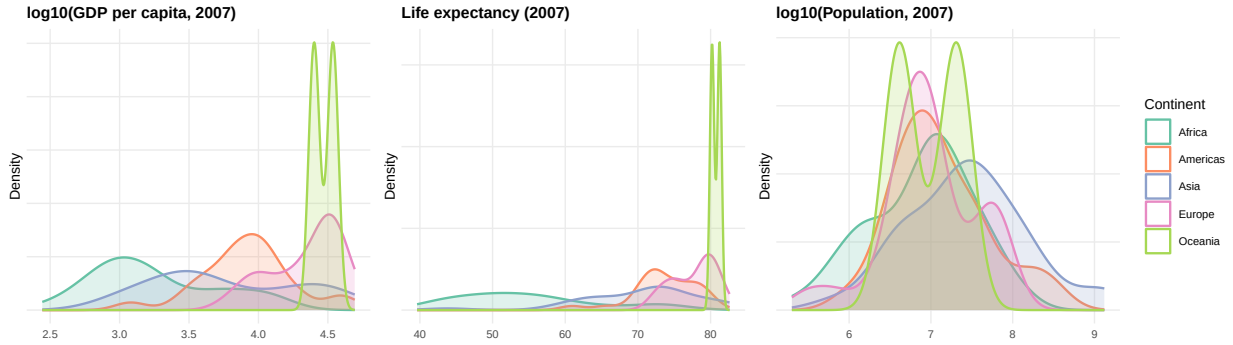

Figure 3: Kernel density estimates of each variable in 2007.

The KDEs show the within-continent distributional shape. Asia's GDP distribution is visibly bimodal (Gulf states and East Asian tigers on one side, South and Central Asia on the other), with an IQR ratio of 8.9, the largest of any continent. Africa's life-expectancy distribution is wide and left-skewed, with a lower tail formed by Swaziland, Zambia, Zimbabwe, and Lesotho, all below 45 years. Population densities peak near 10 million in every continent, with Asia carrying the long right tail that contains China and India. Continents that appeared internally homogeneous on the trajectory plot thus conceal real heterogeneity under the mean.

Table 1: Median GDP per capita (USD) and life expectancy (years) by continent; population (millions).

	N	GDP 1952	GDP 2007	LE 1952	LE 2007	Pop 2007 (M)
Africa	52	987	1452	38.8	52.9	930
Americas	25	3048	8948	54.7	72.9	899
Asia	32	1148	4328	44.4	72.2	3809
Europe	30	5142	28054	65.9	78.6	586
Oceania	2	10298	29810	69.3	80.7	25

A small number of countries deviate strongly from their continent's trend and reappear as extreme points in the PCA, CCA, and MDS projections below. Equatorial Guinea's GDP per capita rose 32× on the back of offshore oil discoveries, Zimbabwe's life expectancy fell by five years, Singapore, South Korea, and Taiwan achieved sustained 20–24× GDP multiples, and the Democratic Republic of the Congo and Haiti are the two economies that regressed most sharply, ending the period at 36% and 65% of their 1952 GDP per capita respectively.

1.2 Principal Component Analysis

We perform PCA independently on the 141×12 GDP and life-expectancy blocks. GDP values span more than two orders of magnitude across countries, so we standardise each year column (which is equivalent to decomposing the correlation matrix) to prevent larger recent-year variances from dominating. Life expectancy is on a comparable scale across years, but we also standardise it. The covariance and correlation decompositions produce near-identical results (PC1 explains 92.0% and 92.3% of the variance respectively), and standardising keeps the treatment of the two variable blocks consistent.

1.2.1 Scree analysis and variance explained

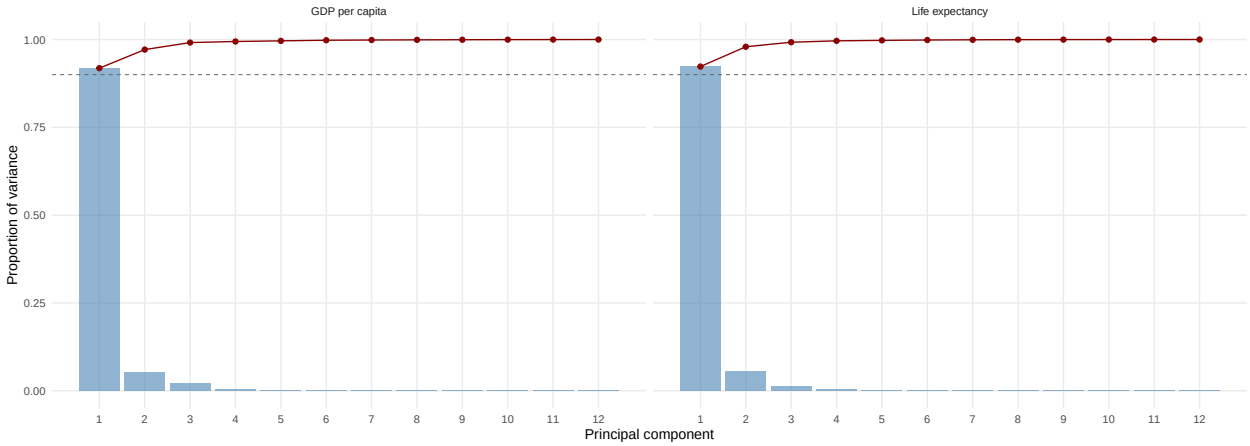

Figure 4: Scree plots for the GDP and life-expectancy PCAs.

Both decompositions concentrate almost all variance in a single component. PC1 explains 91.8% of the GDP variance and 92.3% of the life-expectancy variance, while PC2 adds a further 5.3% and 5.6% respectively. Two components are sufficient, together covering 97.1% of the GDP variance and 97.9% of the life-expectancy variance. This follows directly from the year-to-year redundancy within each block, with adjacent years correlated at approximately 0.99, so 12 columns compress cleanly into two.

1.2.2 Interpretation of PC1 and PC2

The loadings have the same structure on both decompositions. PC1 assigns nearly uniform positive weights across all 12 years (0.28 to 0.30 on GDP, 0.27 to 0.30 on life expectancy), so a country's PC1 score is approximately proportional to its average level over the full 55-year period. For PC2, the loadings begin strongly negative in 1952 (-0.35) and rise monotonically to strongly positive in 2007 ($+0.42$). A positive PC2 score therefore indicates a country whose values were low early and high late (a catching-up or fast-growth trajectory), and a negative PC2 score indicates the opposite (stagnation or decline from a relatively high base).

1.2.3 PC score scatter plots

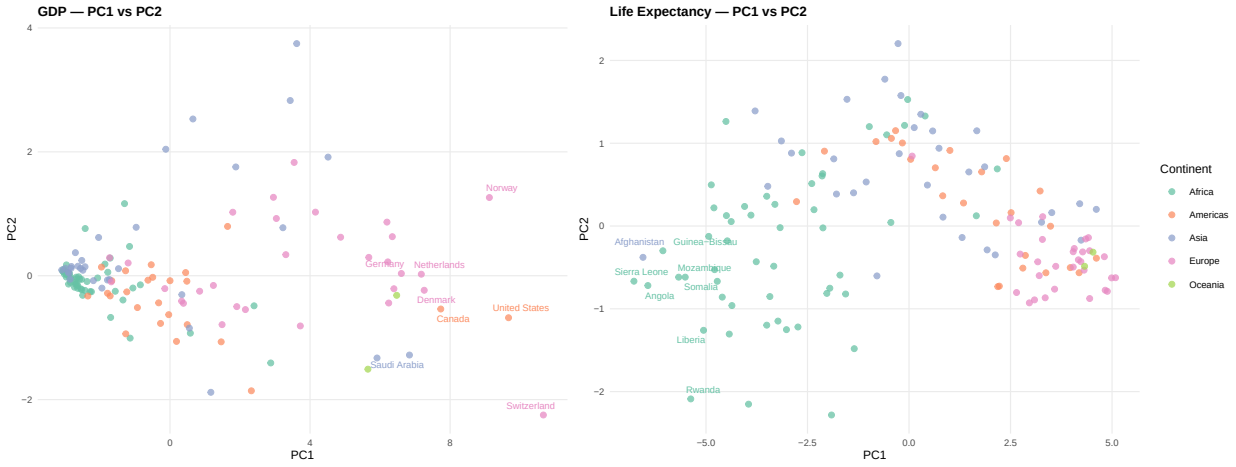

Figure 5: PC1 vs PC2 scatter plots for GDP (left) and life expectancy (right), coloured by continent.

The GDP scatter is structured by level on PC1. Switzerland, the US, Norway, Canada, and Denmark sit at the top end, while Myanmar, Burundi, Ethiopia, Eritrea, and Malawi sit at the bottom. PC2 then separates countries by growth trajectory. The top Asian countries form the obvious cluster at the top of PC2 (Singapore, Hong Kong, Taiwan, South Korea, Japan), identifying the countries whose relative position improved most over the period. Strongly negative PC2 scores belong to Venezuela, Iraq, and Libya, relatively high-income economies in 1952 that did not sustain their early position advantage.

The life-expectancy scatter has the same geometry. PC1 separates Iceland, Sweden, Norway, the Netherlands, and Switzerland at the top and Sierra Leone, Afghanistan, Angola, Guinea-Bissau, and Mozambique at the bottom. PC2 captures the rate of improvement, with Oman, Vietnam, Saudi Arabia, Indonesia, and Libya gaining life expectancy fastest, while Zimbabwe, Zambia, Rwanda, Botswana, and the Democratic Republic of the Congo have the strongly negative PC2 scores consistent with the health and conflict downturn identified in the EDA. Continent clusters are distinct, with Europe and Oceania at the top of PC1, Africa at the bottom, the Americas in the middle, and Asia stretched along PC2 consistent with its bimodal GDP distribution.

1.2.4 GDP PC1 versus life-expectancy PC1

Figure 6: First PC scores, GDP against life expectancy.

The two PC1 scores plotted against each other show how economic level and health level covary over the full period. The Pearson correlation is 0.77, consistent with the 0.81 log(GDP)–life expectancy correlation seen in the 2007 snapshot and confirming the long-run coupling between wealth and health. Most countries lie close to the diagonal. The systematic deviations are countries whose health outcomes are either stronger or weaker than their income level would predict, reflecting external factors such as public health investment, diet, and health system access that the economic dimension alone does not capture.

1.2.5 Limitations of linear PCA

The decomposition is restricted to linear combinations of the year columns and the GDP–life expectancy relationship is non-linear (Pearson correlation 0.68 on raw GDP against 0.81 on log(GDP)). Kernel PCA, which applies the standard decomposition in a feature space induced by a kernel, would capture this curvature directly and would be expected to produce tighter continent separation than

the linear projection used here.[1]

1.3 Canonical Correlation Analysis

We apply CCA with log(GDP) as the first variable set and life expectancy as the second, giving $p = q = 12$. The log transform is motivated by the same skewness and non-linearity noted in the EDA and the PCA limitations subsection. Raw GDP is strongly right-skewed (skewness 1.88) and the 2007 GDP–life expectancy relationship curves noticeably on the raw scale but is close to linear on the log scale.

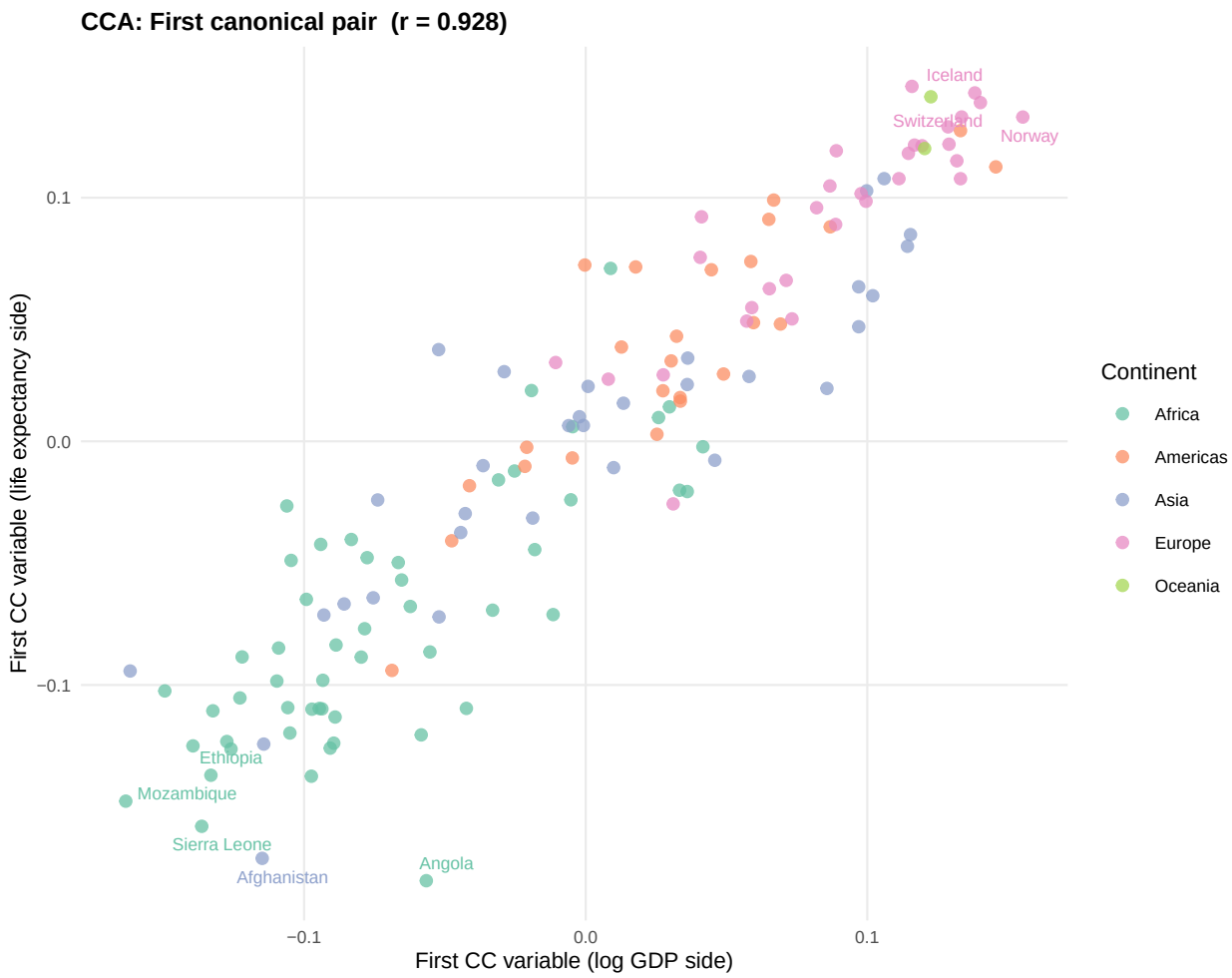

Figure 7: First pair of canonical variables, coloured by continent.

1.3.1 Canonical correlations

The first canonical correlation is 0.93, and subsequent correlations decay to 0.63, 0.54, and 0.5, with the remainder below 0.40. The first pair therefore carries most of the shared linear structure between the two blocks. Looking at CC2 directly, the GDP-side loadings start near zero in 1952 (0.003), rise to 0.11 around 1977, and fall back to −0.11 by 2007, while the life-expectancy loadings are uniformly small and negative across all 12 years (−0.05 to −0.19). The GDP pattern alone describes a faint

[1] Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning* (2nd ed., 2009), Chapter 14.

mid-period growth contrast, but it is not matched by any equivalent temporal structure on the life-expectancy side, so the two variates of CC2 are not aligned on the same temporal feature and the pair has no joint interpretation. Higher-order pairs decay further and are similarly uninterpretable.

1.3.2 Interpretation of the first canonical pair

The first canonical axis ranks countries by a joint development level that log(GDP) and life expectancy identify together. The lower extreme is dominated by sub-Saharan African and conflict-affected Asian states, with Mozambique, Myanmar, and Sierra Leone at the bottom, and the upper extreme by Western European countries and the United States, with Norway and Switzerland at the top. This ordering is near-identical to the one PCA produced on either variable separately, because the 12 year columns within each block are correlated at approximately 0.99 and there is little room for CCA's maximum-correlation objective to choose a direction meaningfully different from PCA's maximum-variance one. The continent structure on the canonical scatter reproduces the PCA pattern, with Africa in the lower-left and Europe and Oceania in the upper-right.

1.3.3 Effect of the log transform

Repeating the analysis on raw GDP gives a first canonical correlation of 0.89, materially lower than the 0.93 achieved with log(GDP). CCA only finds linear combinations of the variables it is given, so any curvature in the raw GDP–life expectancy relationship is left unaccounted for. The log transform straightens the relationship and raises the canonical correlation. The gain is the PCA-section's linearity limitation measured on a different axis.

1.4 Multidimensional Scaling

Classical (metric) MDS is applied to the combined 36-dimensional dataset formed by concatenating log(GDP), life expectancy, and log(population), using Euclidean distance as specified in the coursework. The log transforms on GDP and population bring them onto a roughly comparable scale with life expectancy.

The Pearson correlation between pairwise distances in the original 36-D space and the 2D MDS projection is 0.998, so the two retained dimensions preserve the distance geometry of the full data almost exactly.

Figure 8: Two-dimensional MDS representation, coloured by continent.

1.4.1 Development axis

Dimension 1 is a clear development axis. The continent centroids line up from Africa at -37 through the Americas at $+18$ and Europe at $+44$ to Oceania at $+52$, in the same order they occupy on every previous projection. The lower extreme is dominated by sub-Saharan African and conflict-affected Asian states such as Sierra Leone, Afghanistan, Angola, Guinea-Bissau, and Rwanda, while the upper extreme is occupied by Scandinavian and Alpine European countries including Sweden, Iceland, Norway, the Netherlands, and Switzerland. This country ordering matches the one produced by PCA and CCA, since all three methods are projecting the same dominant development signal that accounts for the bulk of the variation in the data.

1.4.2 Growth direction

Dimension 2 separates countries by the direction of change in life expectancy over the period rather than by level. The most negative scores belong to countries with the largest life-expectancy gains over the 55 years, a cluster containing Oman, Vietnam, Indonesia, and Yemen, while the most positive scores belong to countries whose life expectancy fell or stagnated over this period, including Zimbabwe, Botswana, Zambia, and Swaziland. Asia's centroid is pulled below the horizontal axis by its concentration of fast-gaining countries, while Africa's centroid sits near the horizontal because

the continent contains both rapidly improving and declining cases. The contrast is the same one that appeared as PC2 in the life-expectancy PCA, now reproduced with all three variables analysed simultaneously.

1.4.3 Relationship to PCA

The agreement between the MDS and PCA axes is not a coincidence. Classical MDS with Euclidean distance is, up to sign, the same decomposition as PCA on the same centered data. Writing the SVD of the centered data matrix as

$$X = UDV^T,$$

PCA returns the scores

$$T_{\mathrm{PCA}} = XV = UD.$$

Classical MDS forms the Gram matrix $B = XX^T$ and takes the eigendecomposition

$$B = UD^2U^T,$$

from which the MDS coordinates are the leading columns of UD, the same matrix as T_{PCA}. The MDS scatter does not literally reproduce either PCA scatter because the input differs. Here MDS is applied to the 36-column concatenation of log(GDP), life expectancy, and log(popn), whereas the PCA sections ran the decomposition on the GDP and life-expectancy blocks separately.

1.4.4 Comparison with the earlier analyses

The MDS projection recovers the same two axes as the individual PCAs, with dimension 1 tracking development level and dimension 2 tracking growth direction. The MDS version adds a third variable by operating on all three simultaneously, so a single 2D summary captures joint structure that required two separate PCAs above. The 0.997 correlation between pairwise distances in the original 36-D space and the 2D MDS projection makes the reduced representation a near-lossless summary, so the development and growth-direction axes are the two dominant sources of variation in the full UN data.

2 Question 2

2.1 Linear Discriminant Analysis

We use LDA to predict continent from all 36 features (GDP, life expectancy, and population across 12 years). The data are split 70/30 into training and test sets.

A single fixed split with seed 42 gives a test accuracy of 0.643 on 42 held-out countries, and averaging over 50 random splits gives a more stable estimate of 0.616 ± 0.074. Both figures sit comfortably above the 0.20 five-class chance rate and above the 0.37 majority-class baseline set by Africa's sample proportion, so the classifier is extracting genuine structure from the features.

Table 2: Rows are true continents, columns are predicted.

	Africa	Americas	Asia	Europe	Oceania
Africa	13	0	1	0	0
Americas	2	2	3	0	1
Asia	2	2	5	1	0
Europe	0	0	1	7	1
Oceania	0	0	0	1	0

Per-class accuracy varies substantially. Africa is the best-classified continent, followed by Europe, Asia, and the Americas. The Oceanian country in the test set is misclassified as European. Asia's errors are spread across Africa, the Americas, and Europe in roughly equal proportions.

The per-class pattern is consistent with LDA's modelling assumptions. The classifier assigns country i to the class c that maximises

$$\delta_c(x) = x^\top \Sigma^{-1} \mu_c - \tfrac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \log \pi_c,$$

where μ_c is the class mean, Σ is the pooled within-class covariance, and π_c is the class prior. The decision rule is therefore linear in x, and the two modelling assumptions behind it are that each class is multivariate normal and that all classes share the same covariance Σ. Both assumptions break on this dataset. The EDA showed Asia's GDP distribution is strongly bimodal, so Asia's single mean μ_{Asia} lands between its two modes and the density $N(\mu_{\text{Asia}}, \Sigma)$ concentrates mass in a region where almost no Asian country actually sits, while overlapping with the Gaussian densities for Africa, the Americas, and Europe. The shared covariance is similarly misspecified because Oceania and Europe have far tighter within-class spread than Africa or Asia, so Σ is an average that fits neither. Africa is classified more accurately than Asia despite being internally heterogeneous because its spread runs along a single unimodal axis, which the Gaussian model can represent even if the variance is wrong.

2.2 Clustering

Three methods from the module are applied to the standardised GDP and life-expectancy blocks (24 features). k-means and the model-based clustering are fitted directly, and hierarchical clustering is applied via its linkage-distance dendrogram, using Ward's linkage as in the computer lab sessions. Ward merges the pair of clusters producing the smallest increase in within-cluster variance at each step. k is chosen from the dendrogram and the MVN-mixture BIC curve, with the same k then used for k-means so that all three methods can be compared on equal footing.

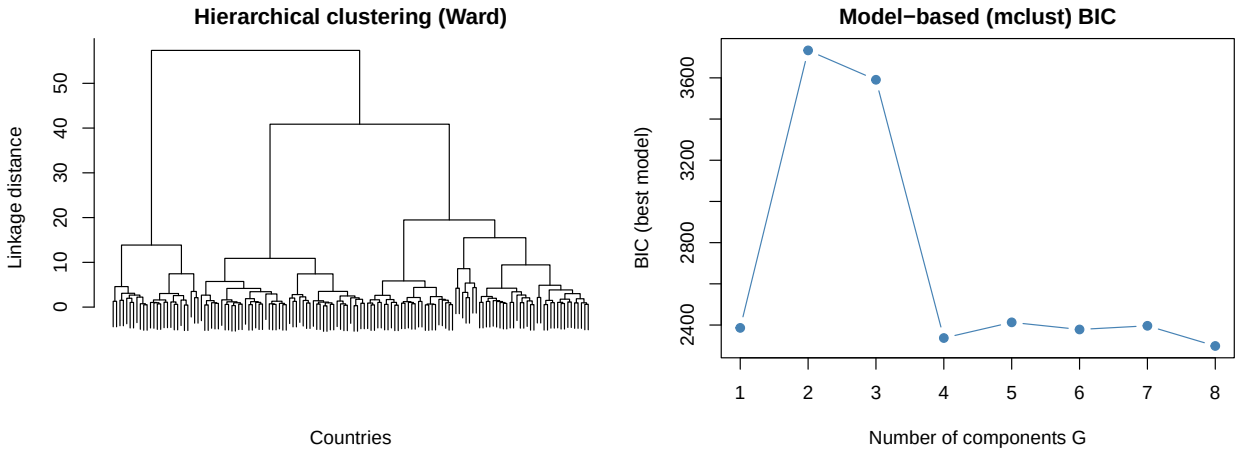

Figure 9: Hierarchical clustering dendrogram (left) and model-based BIC as a function of number of components (right).

2.2.1 Choice of k

The dendrogram's three largest merge distances are 57.4, 40.9, and 19.5, with every subsequent merge below 16. The substantial gap between merges 3 and 4 identifies $k = 3$ as the cleanest cut point from the hierarchical perspective. The MVN-mixture BIC curve has a clear maximum at $G = 2$ (BIC 3733), with $G = 3$ second-best at BIC 3591 and $G \geq 4$ falling sharply. The two selection criteria therefore disagree, with the dendrogram preferring $k = 3$ and the mixture model preferring $k = 2$. We proceed with $k = 3$ because the BIC gap between $G = 2$ and $G = 3$ is moderate relative to the drop-off from $G = 3$ to $G = 4$, and the three-cluster partition is the more informative one against the development-axis structure recovered by PCA and MDS.

2.2.2 Agreement between methods

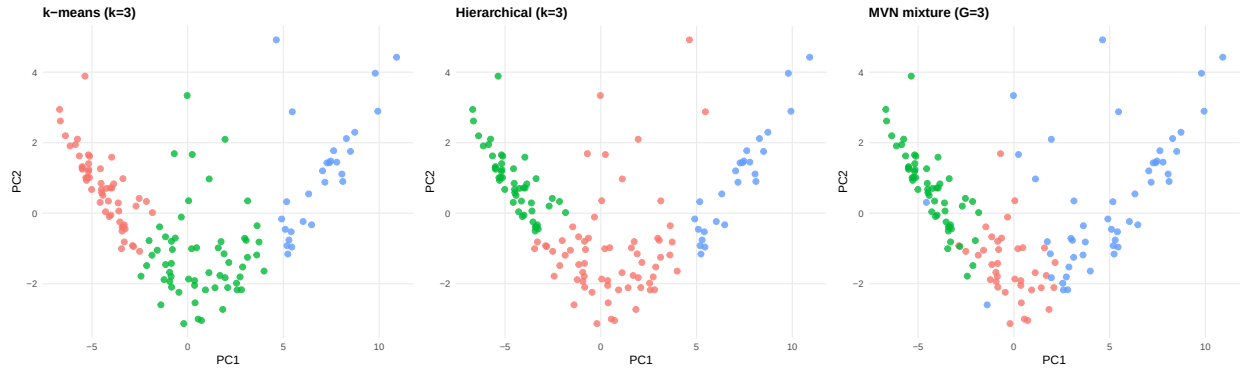

Figure 10: Cluster assignments at k = 3 for k-means, hierarchical clustering, and the MVN mixture.

At $k = 3$ the Adjusted Rand Index is 0.84 between k-means and hierarchical clustering, 0.57 between k-means and the MVN mixture, and 0.51 between hierarchical and the MVN mixture. The k-means and hierarchical results are nearly identical, consistent with both being variance-minimising methods on the same standardised features. The MVN mixture disagrees more because mclust fits

full-rank covariance structures (varying shape, equal volume) rather than the spherical within-cluster geometry that k-means and Ward's method assume, and the covariance flexibility lets it draw cluster boundaries that follow the anisotropic spread of the data rather than minimising within-cluster variance.

2.2.3 Cluster interpretation

The three clusters have a natural development interpretation. The first cluster contains 18 of the 30 European countries, 6 Asian countries, and both Oceanian countries, placing it at the high-GDP and high-life-expectancy end of the development axis already seen in PCA and MDS. The second cluster contains 42 of the 52 African countries together with 10 Asian and 2 American ones, and sits at the low-development end. The third is the middle cluster, dominated by the Americas (21 of 25) with a substantial Asian contingent (16 of 32) and the remaining 12 European countries. The ordering reproduces the PCA and MDS development axis and shows that continent is a reasonable but imperfect proxy for development level. Africa and Europe fall almost entirely into the low and high clusters respectively, while the Americas and Asia are split across the middle and the extremes.

2.3 Linear Regression

We predict 2007 life expectancy from the 11 GDP columns for 1952–2002, excluding 2007 GDP to avoid contemporaneous leakage. The 11 predictors are correlated with one another at approximately 0.99 (the year-to-year structure noted in the EDA), which is textbook multicollinearity and motivates testing methods that handle it directly rather than OLS alone. Six models are compared. Ordinary least squares, ridge regression, and principal component regression (PCR) are each applied to raw GDP and to log(GDP).

Performance is measured by leave-one-out cross-validation (LOOCV). Each country is held out in turn, the model is refit on the remaining 140, and the held-out country is predicted from that fit.[2] Writing $\hat{y}_i$ for the prediction made for country i by the model fit without it, the LOOCV score is

$$\mathrm{RMSE}_{\mathrm{LOOCV}} = \sqrt{\frac{1}{141}\sum_{i=1}^{141}(y_i - \hat{y}_i)^2}.$$

Each $\hat{y}_i$ is produced by a fit that never saw country i, so the 141 residuals are genuine prediction errors rather than in-sample fit.

Ridge and PCR have hyperparameters (α and k), which we tune by nested cross-validation. The LOOCV above is the outer loop. Within each outer fold, an inner 10-fold CV on the 140 training rows picks α or k, and the selected value is used to refit on those 140 rows and predict the held-out country. The held-out country never enters the inner loop, so it cannot influence the hyperparameter used to score it. A single flat LOOCV that chose α or k from all 141 residuals would feed each country's residual back into its own hyperparameter selection, pulling the reported RMSE below the true out-of-sample value. OLS has no hyperparameter and is unaffected.

2.3.1 Ridge regression and principal component regression

Ridge regression minimises

$$\|y - X\beta\|^2 + \alpha\|\beta\|^2,$$

[2] Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning* (2nd ed., 2009), Section 7.10.

adding a penalty on the squared length of β to the usual least-squares loss. When predictors are near-collinear, OLS fits tend to pair large positive coefficients with large negative ones that mostly cancel on the fitted values. The ridge penalty rules these out and returns a smaller, more stable set of coefficients in exchange for a small amount of bias. The parameter α sets the trade-off. At $\alpha = 0$ Ridge reduces to OLS, and as $\alpha \to \infty$ the coefficients are driven to zero. We search α on a 50-point log-spaced grid in $[10^{-2}, 10^{3}]$.

Principal component regression replaces the 11 near-collinear year columns with their first k principal components and fits OLS in that reduced space. The PC directions beyond the leading few carry almost no variance in the GDP block, so including them in OLS would inflate the fit's sensitivity to them without supplying any real signal. Cutting the number of components at k eliminates those directions outright.

2.3.2 Ridge and PCR as shrinkage on principal directions

Both estimators can be written as filters on the principal directions of the predictor matrix. Let the SVD of the centered X be

$$X = UDV^T,$$

where D holds the singular values in decreasing order. The OLS fit projects y onto the column space of X, weighting every PC direction equally, while Ridge and PCR replace those unit weights with component-wise filters,

$$\hat{y}_{\text{OLS}} = \sum_i u_i u_i^T y, \qquad \hat{y}_{\text{Ridge}} = \sum_i \frac{d_i^2}{d_i^2 + \alpha} u_i u_i^T y, \qquad \hat{y}_{\text{PCR}} = \sum_{i=1}^{k} u_i u_i^T y.$$

Ridge's factor is nearly one on high-variance directions and near zero on low-variance ones, giving a smooth shrinkage. PCR is the hard version, keeping the first k directions untouched and dropping the rest. In the GDP block the first three singular values dominate the remaining eight by orders of magnitude, so both filters produce fits built almost entirely from the first three PC directions. Ridge's nested-CV RMSE and PCR's are two smoothings of the same low-rank structure, explaining why they land so close.

On log(GDP) the selected α is stable across the 141 outer folds (range 3.6–11.5, median 7.2). On raw GDP it is less stable (range 0.01–47.1, median 23.3), reflecting a flat cross-validation surface on the raw scale where different subsets of 140 countries favour very different penalties.

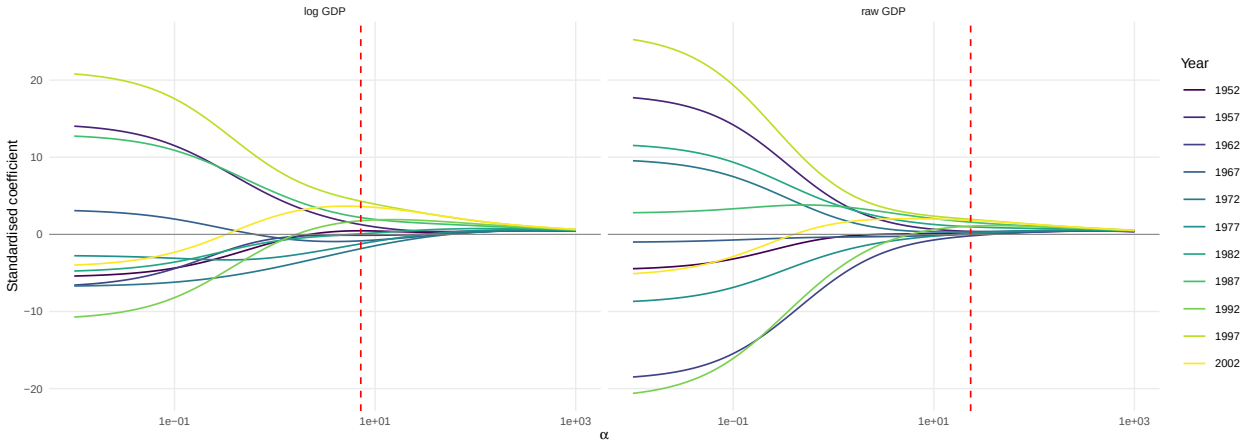

Figure 11: Standardised Ridge coefficients as a function of the penalty α, on raw GDP (left) and log GDP (right). The vertical dashed line marks the median α selected across the 141 nested-LOOCV outer folds.

At low α the eleven year coefficients are large and opposing in sign. This is expected when predictors are near-collinear, since OLS can produce many combinations of large coefficients that mostly cancel when predicting the response. Increasing α shrinks the coefficients toward zero and flattens the year-to-year differences between them. On log(GDP) the nested-CV median sits in the intermediate region, where shrinkage has stabilised the coefficients without yet removing their signal.

Table 3: Nested leave-one-out cross-validated performance.

Model	LOOCV RMSE	LOOCV R2
OLS (raw GDP)	9.26	0.41
OLS (log GDP)	7.64	0.60
Ridge (raw GDP)	9.04	0.44
Ridge (log GDP)	7.25	0.64
PCR (raw GDP, k=2)	8.97	0.44
PCR (log GDP, k=3)	7.15	0.65

OLS on raw GDP gives a LOOCV RMSE of 9.26 years and $R^2 = 0.41$, while OLS on log(GDP) gives RMSE 7.64 and $R^2 = 0.60$, a one-and-a-half-year reduction in prediction error from the transform. Ridge on log(GDP) improves this further to RMSE 7.25 and $R^2 = 0.64$, while Ridge on raw GDP reaches RMSE 9.04 and $R^2 = 0.44$, a modest gain over the raw-scale OLS. GDP's skewness makes the linear relationship with life expectancy non-linear on the raw scale but close to linear on the log scale, and the regression results express the cost of ignoring this as 0.19 units of R^2.

PCR has the lowest RMSE of the six models. On log(GDP) the inner CV selected $k = 3$ components in every one of the 141 outer folds, and the resulting nested LOOCV RMSE is 7.15 with $R^2 = 0.65$. This stability is diagnostic of the underlying structure. The PCA of the GDP block showed that the first three components already explain 99% of the variance, so removing any single country never changes which directions dominate the design matrix and the inner CV reaches the same conclusion every time. PCR on raw GDP peaks at $k = 2$ (modal selection) with RMSE 8.97, but

still does not reach the log-based models, confirming that the transform is more important than the multicollinearity fix.

The nested design lets us separate two effects. For PCR the held-out country never influences the k used to score it (because k is constant across folds), so the nested RMSE is an unbiased estimate of out-of-sample RMSE. For Ridge the inner-selected α does vary across outer folds, so the nested score exceeds what a single flat LOOCV score would suggest. PCR's zero tuning variance makes it the preferred model of the two. Both outperform OLS on log(GDP) because they address the near-collinearity of the year columns, with PCR doing so more cleanly.

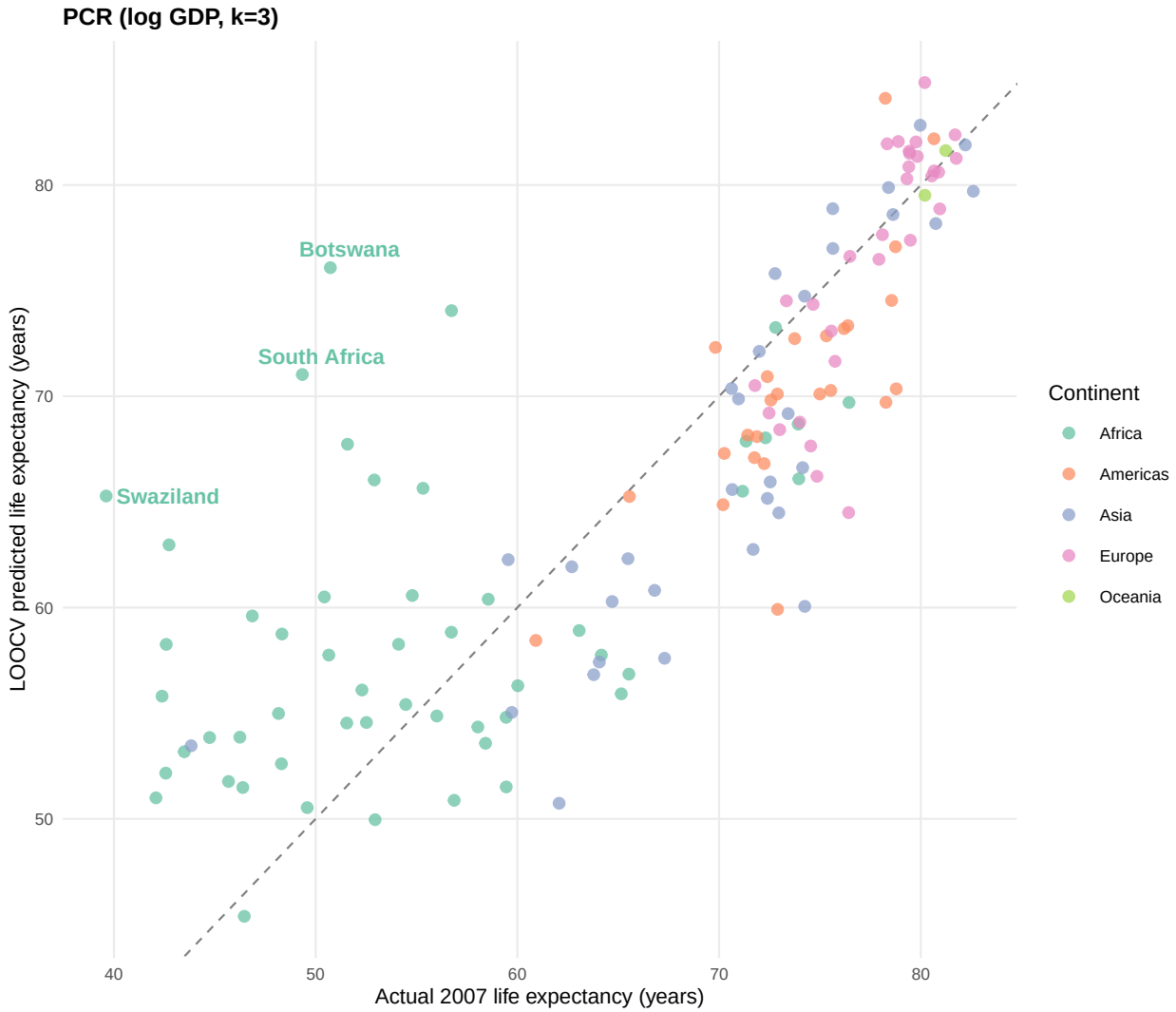

Figure 12: Leave-one-out predictions against actual 2007 life expectancy for the best model.

The residuals have a clear structure. The standard deviation of the 141 leave-one-out residuals is 7.17, and 7 countries sit beyond two standard deviations of the fit. All seven are African and all are over-predicted. Swaziland, Botswana, and South Africa lie above the diagonal by more than 20 years, with Angola, Lesotho, Equatorial Guinea, and Gabon between 15 and 17 years. Zambia, Namibia, and Nigeria sit just inside the two-SD threshold, extending the same pattern. The direction

is systematic rather than random. Countries with pre-1990 GDP histories comparable to other middle-income nations experienced two shocks the model cannot see from GDP alone. The first is a health cluster in southern Africa (Swaziland, Botswana, Lesotho, South Africa, Zambia, Namibia), where life expectancy fell between 1990 and 2007 without a proportional fall in income. The second is an oil-rent pattern (Equatorial Guinea, Gabon, Angola), where per-capita GDP was pushed up by resource extraction concentrated in a narrow elite without matching gains in public health. The two negative-residual outliers are Vietnam (-14.2 years, life expectancy outpaces GDP) and Nicaragua (-13.0), which show the opposite gap. A health shock that leaves GDP largely unaffected, or a health gain decoupled from GDP, is invisible to the model, and these countries sit furthest from the diagonal under every combination of transform, regularisation, and component count we tried.